

Engineering Mathematics Study Notes: Geometric Series Test

1. Concept Identification

The notes introduce the **Geometric Series Test**, which is a direct, inspection-based test used to determine the convergence or divergence of an infinite series where each consecutive term is found by multiplying the previous term by a fixed, constant multiplier called the **common ratio**.

2. Mathematical Representations

Standard Form of a Geometric Series

An infinite geometric series is written algebraically as:

$$\sum_{n=0}^{\infty} a \cdot r^n = a + ar + ar^2 + ar^3 + \dots + ar^n + \dots$$

Where:

- a is the first term of the series ($a \neq 0$).
- r is the constant common ratio between consecutive terms.

The Geometric Series Convergence Criteria

The absolute value of the common ratio, denoted as $|r|$, completely dictates the behavior of the infinite sum:

If $|r| < 1 \implies$ The series is Convergent

If $|r| \geq 1 \implies$ The series is Divergent

Sum of an Infinite Convergent Geometric Series

If a geometric series satisfies the convergence condition ($|r| < 1$), its total infinite summation value settles exactly on a finite value calculated by:

$$S_{\infty} = \frac{a}{1 - r}$$

3. The Algorithm (Procedural Recipe)

To evaluate an infinite series using the Geometric Series framework from absolute first principles, execute these steps mechanically:

Step 1: Extract Individual Terms

Expand the summation notation to list out the first three or four sequential terms of the numerical series explicitly.

Step 2: Determine the First Term (a)

Identify the very first numeric term in the expanded progression and assign it to the variable a .

$$a = \text{Term 1}$$

Step 3: Compute the Common Ratio (r)

Divide any term by its immediate predecessor to find the constant multiplier value r :

$$r = \frac{\text{Term 2}}{\text{Term 1}} = \frac{\text{Term 3}}{\text{Term 2}}$$

Step 4: Apply the Absolute Value Condition

Find the positive magnitude of the common ratio ($|r|$) and check it against the threshold value of 1.

Step 5: Classify the Convergence Behavior

- If the absolute value is strictly less than 1, write **"Convergent"** and proceed to Step 6.
- If the absolute value is greater than or equal to 1, write **"Divergent"** and terminate the procedure.

Step 6: Compute the Final Sum (Only if Convergent)

Substitute the values of a and r into the infinite sum formula to resolve the final numeric value of the entire infinite addition.

4. Operational Mechanics

Mechanical Example 1: Solving a Fractional Decreasing Geometric Series

Let's solve the primary example presented across the lecture boards:

$$\text{Test the series: } 1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \infty$$

Step 1: Extract parameter a

Look at the beginning of the series expression:

$$a = 1$$

Step 2: Calculate parameter r

Divide the second term by the first term:

$$r = \frac{\frac{1}{3}}{1} = \frac{1}{3}$$

Let's verify this remains constant by checking the next pair:

$$r = \frac{\frac{1}{9}}{\frac{1}{3}} = \frac{1}{9} \times \frac{3}{1} = \frac{3}{9} = \frac{1}{3}$$

The common ratio is verified mechanically as $r = \frac{1}{3}$.

Step 3: Test the stability condition

Take the absolute value of our ratio:

$$|r| = \left| \frac{1}{3} \right| = \frac{1}{3}$$

Compare this fraction directly against 1:

$$\frac{1}{3} < 1 \implies \text{The series is Convergent}$$

Step 4: Compute the final infinite sum

Since the series is confirmed to be convergent, compute its exact total final value:

$$S_{\infty} = \frac{1}{1 - \frac{1}{3}}$$

Simplify the denominator expression ($1 - \frac{1}{3} = \frac{2}{3}$):

$$S_{\infty} = \frac{1}{\frac{2}{3}}$$

Multiply by the reciprocal of the denominator fraction to find the final evaluation:

$$S_{\infty} = 1 \times \frac{3}{2} = \frac{3}{2}$$

Mechanical Example 2: Recognizing a Divergent Geometric Series

Consider a series where terms grow larger over time:

Test the series: $2 + 4 + 8 + 16 + 32 + \dots \infty$

Step 1: Extract parameter a

$$a = 2$$

Step 2: Calculate parameter r

Divide the second term by the first term:

$$r = \frac{4}{2} = 2$$

Step 3: Test the stability condition

Take the absolute value of our ratio:

$$|r| = |2| = 2$$

Compare this value directly against the divergent threshold:

$$2 \geq 1 \implies \text{The series is Divergent}$$

The terms grow infinitely large, meaning the total sum escapes to infinity. No summation calculation can be performed.